



## DEPARTMENT OF INFORMATION TECHNOLOGY

# FACULTY PROFILE REPORT

Employee Code: 7001 | Generated: 29-04-2026

### 1. PERSONAL INFORMATION

|                   |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|
| Full Name         | KAMBHAMPATI VIJAY KUMAR                                                              |
| Employee Code     | 7001                                                                                 |
| Father's Name     | KAMBHAMPATI KANAKA DURGA RAO                                                         |
| Mother's Name     | KAMBHAMPATI SHOBHANA KUMARI                                                          |
| Date of Birth     | 24 May 1984                                                                          |
| Gender            | Male                                                                                 |
| State             | Telangana                                                                            |
| Caste             | BC-D                                                                                 |
| Sub-Caste         | MUNNURUKAPU                                                                          |
| Nationality       | Indian                                                                               |
| Permanent Address | H.No:- 6-35/12, MARUTHI NAGAR, WARANGAL CROSS ROAD,<br>YEDULAPURAM, KHAMMAM - 507163 |

### 2. PROFESSIONAL INFORMATION

|                      |                           |
|----------------------|---------------------------|
| Department           | IT                        |
| Designation          | Assistant Professor       |
| Joining Date         | 10 Dec 2010               |
| Experience at Anurag | 15 Years 3 Months         |
| Experience (Other)   | 1                         |
| Total Experience     | 15 Years 4 Months 19 Days |
| Email                | hod.inf@anurag.ac.in      |
| Mobile               | 9553122276                |
| Phone                | 9030900962                |
| Current Status       | ACTIVE                    |

### 3. PROFESSIONAL IDs

|                      |                     |
|----------------------|---------------------|
| <b>JNTUH ID</b>      | 15150406-114529     |
| <b>AICTE ID</b>      | 1-468376653         |
| <b>PAN Number</b>    | ARXPK1671R          |
| <b>Aadhar Number</b> | 846452283847        |
| <b>APAAR ID</b>      | 72-265-370-947      |
| <b>ORCID ID</b>      | 0000-0002-9791-8445 |

#### 4. EDUCATIONAL QUALIFICATIONS - SSC / 10TH

|                   |                                       |
|-------------------|---------------------------------------|
| <b>Year</b>       | 2000                                  |
| <b>Percentage</b> | 56.2                                  |
| <b>School</b>     | SES VV CENTRAL PUBLIC SCHOOL, KHAMMAM |

#### 5. EDUCATIONAL QUALIFICATIONS - INTERMEDIATE

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Year</b>       | 2002                              |
| <b>Percentage</b> | 57.1                              |
| <b>College</b>    | EXCELLENT JUNIOR COLLEGE, KHAMMAM |

#### 6. EDUCATIONAL QUALIFICATIONS - UG

|                       |                                           |
|-----------------------|-------------------------------------------|
| <b>Degree</b>         | B.TECH                                    |
| <b>Year</b>           | 2006                                      |
| <b>Percentage</b>     | 55.6                                      |
| <b>College</b>        | SREE KAVITHA ENGINEERING COLLEGE, KHAMMAM |
| <b>Specialization</b> | INFORMATION TECHNOLOGY                    |

#### 7. EDUCATIONAL QUALIFICATIONS - PG

|                       |                                           |
|-----------------------|-------------------------------------------|
| <b>Degree</b>         | M.TECH                                    |
| <b>Year</b>           | 2010                                      |
| <b>Percentage</b>     | 72.4                                      |
| <b>College</b>        | SREE KAVITHA ENGINEERING COLLEGE, KHAMMAM |
| <b>Specialization</b> | SOFTWARE ENGINEERING                      |

#### 8. EDUCATIONAL QUALIFICATIONS - PhD

|               |          |
|---------------|----------|
| <b>Status</b> | Pursuing |
|---------------|----------|

|                |                                  |
|----------------|----------------------------------|
| Year           | Not Specified                    |
| University     | KLU, VIJAYAWADA                  |
| Specialization | COMPUTER SCIENCE AND ENGINEERING |

9. SUBJECTS DEALT

DBMS

PPS

DATA MINING

10. SCM (SERVICE/CONSULTANCY/MOU)

K. VIJAY KUMAR SCM ON 28-12-2019

11. RESEARCH & PUBLICATIONS

| Type            | Title                                                                                    | Academic Year | Year | Status    | Details                                                                                                                                                                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------|---------------|------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Journal Article | Analytical Framework to Quantify and Scale Air Pollution Intensity with Machine Learning | 2024-2025     | 2024 | Published | <b>Authors:</b> K. VIJAY KUMAR & Dr.V.M. SUBRAMANYAM<br><b>Journal:</b> AFRICAN JOURNAL OF BIOLOGICAL SCIENCES<br><b>ISSN:</b> 2663-2187<br><b>DOI:</b> <a href="https://doi.org/10.33472/AFJBS.6.Si2.2024.1421-1427">https://doi.org/10.33472/AFJBS.6.Si2.2024.1421-1427</a><br><b>URL:</b> <a href="http://www.afjbs.com">http://www.afjbs.com</a> |

13. FDP / WORKSHOPS ATTENDED

| Type | Title                            | Academic Year | Duration                            | Organized By                                                                                                                       | Place     |
|------|----------------------------------|---------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| FDP  | Advances in GenAI and Agentic AI | 2024-2025     | 16 Jun 2025 to 21 Jun 2025 (6 days) | Department of Computer Science and Engineering in collaboration with Computer Society of India, CSI-SB, CVR College of Engineering | Hyderabad |

14. B.TECH PROJECTS HANDLED

| H.T.No.    | Student Name    | Batch | Project Title | Approved | Marks |
|------------|-----------------|-------|---------------|----------|-------|
| 23C11A1213 | DHARANI KAMBALA | A1    | ONLINE EXAM   | Yes      | 45    |

15. RESULTS / ACADEMIC PERFORMANCE

**Database Management Systems (CS404PC)** 2024-2025  
Classes Taken: 64 | Students Attempted: 50 | Passed: 49  
**Pass Percentage: 98.0%**

## 16. UPLOADED DOCUMENTS

**Aadhar Card:** ✓ Uploaded

**PAN Card:** ✓ Uploaded

**APAAR Document:** ✓ Uploaded

**SCM Document:** ✓ Uploaded

**JNTUH Bio-Data:** ✓ Uploaded

**SSC Certificate:** ✓ Uploaded

**Intermediate Certificate:** ✓ Uploaded

**UG Certificate:** ✓ Uploaded

**PG Certificate:** ✓ Uploaded

**PhD Certificate:** ✗ Not Uploaded

## 17. ADDITIONAL DOCUMENTS

**Research Publications Proof:** ✓ 7 pages uploaded AY: 2024-2025

**FDP Proof:** ✓ 1 pages uploaded AY: 2024-2025

**Experience Certificates:** ✗ Not Uploaded

**Other Documents:** ✗ Not Uploaded

**Research Publications Proof (2024-2025):**

|

**FDP Proof (2024-2025):**

|

## 18. ABOUT YOURSELF

I WAS PROMOTED AS HOD OF IT ON 28-12-2023





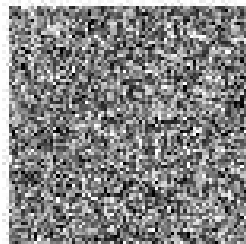


భారత ప్రభుత్వం  
Government of India

భారత విశిష్ట గుర్తింపు ప్రాధికార సంస్థ  
Unique Identification Authority of India

రెజిస్ట్రేషన్/Enrolment No.: 0705/50703/05488

To  
కంభంపాటి విజయ కుమార్  
Kambhampati Vijay Kumar  
S/O Kambhampati Kanaka Durga Rao,  
6-35/12,  
Maruthi Nagar,  
Warangal X Road,  
Khammam rural,  
VTC: Edulapuram (Rural),  
PO: Edulapuram,  
Sub District: Khammam (rural),  
District: Khammam,  
State: Telangana,  
PIN Code: 507163,  
Mobile: 9000900962



మీ ఆధార్ సంఖ్య / Your Aadhaar No. :

8464 5228 3847

VID : 9152 5598 6642 3770

నా ఆధార్, నా గుర్తింపు



భారత ప్రభుత్వం  
Government of India



కంభంపాటి విజయ కుమార్  
Kambhampati Vijay Kumar  
పుట్టిన తేదీ/DOB: 24/05/1984  
ప్రభుత్వం/MALE

ఆధార్ అనేది గుర్తింపు దస్తావ్యం మాత్రమే, పౌరత్వం లేదా పుట్టిన తేదీ కి కాదు. ఇది ధృవీకరించే ద్వారానే అభియంతరించే అంశాలైన ప్రభుత్వం లేదా QR కోడ్ / అన్లైన్ సేవల యొక్క స్వాధీనం.

Aadhaar is proof of identity, not of citizenship or date of birth, it should be used with verification (online authentication, or scanning of QR code / offline NDL).

8464 5228 3847

నా ఆధార్, నా గుర్తింపు



సమాచారము / INFORMATION

- ఆధార్ అనేది గుర్తింపు దస్తావ్యం, పౌరత్వం లేదా పుట్టిన తేదీ కి కాదు. పుట్టిన తేదీ అనేది ఆధార్ సంఖ్య చొప్పున సమన్వయించే సేవలందుకునే ఏర్పాట్ల పుట్టిన తేదీ దస్తావ్యం ద్వారా ఆధారు ద్వారా ఇచ్చే సమాచారంపై ఆధారపడి ఉంటుంది.
- ఈ ఆధార్ లేకపోతే UIDAI వెబ్సైటుపై ప్రమాణీకరించే ఏకస్థితి ద్వారా అన్లైన్ ప్రమాణీకరించే ద్వారా లేదా యాన్ డ్రౌన్ లో అందుబాటులో ఉన్న mAadhaar లేదా ఆధార్ QR స్కానర్ యాప్స్ ఉపయోగించి లేదా [www.uidai.gov.in](http://www.uidai.gov.in) లో అందుబాటులో ఉన్న మరలిన QR కోడ్ కింద యాన్ లైన్ ఉపయోగించి QR కోడ్ స్కాన్ చేసి ద్వారా ధృవీకరించాలి.
- ఆధార్ ప్రశ్నోత్తరాలకు మరలు సులభమైనది.
- ఆధార్ పరిమితి లేనిది తేదీ ముందే పూర్తి 10 సంవత్సరాల తర్వాత గుర్తింపు మరలు రేఖాంశాలకు సంబంధించిన చట్టంలో ఆధార్ ను సవరించాలి.
- వివిధ ప్రభుత్వ మరియు ప్రభుత్వాల ప్రయోజనాలు/సేవలను పొందడంలో ఆధార్ మీకు సహాయపడుతుంది.
- మీ మొత్తం దస్తావ్య మరలు ఈ-మెయిల్ ద్వారా ఆధార్ లో అప్డేట్ చేసుకోండి.
- ఆధార్ సేవలను పొందేందుకు mAadhaar యాప్ ను డౌన్లోడ్ చేసుకోండి.
- ఆధార్/అభియంతరించే ఉపయోగించే దస్తావ్యం ద్వారా విధించబడిన లాక్/అన్లైన్ ఆధార్/అభియంతరించే సేవలకు ఉపయోగించండి.
- ఆధారును కోరి సంఖ్యల ద్వారా విధించే దస్తావ్యం పొందటం.
- Aadhaar is proof of identity, not of citizenship or date of birth (DOB). DOB is based on information supported by proof of DOB document specified in regulations, submitted by Aadhaar number holder.
- This Aadhaar letter should be verified through either online authentication by UIDAI-appointed authentication agency or QR code scanning using mAadhaar or Aadhaar QR Scanner app available in app stores or using secure QR code reader app available on [www.uidai.gov.in](http://www.uidai.gov.in).
- Aadhaar is unique and secure.
- Documents to support identity and address should be updated in Aadhaar after every 10 years from date of enrolment for Aadhaar.
- Aadhaar helps you avail of various Government and Non-Government benefits/services.
- Keep your mobile number and email id updated in Aadhaar.
- Download mAadhaar app to avail of Aadhaar services.
- Use the feature of Lock/Unlock Aadhaar biometrics to ensure security when not using Aadhaar biometrics.
- Entities seeking Aadhaar are obligated to seek consent.

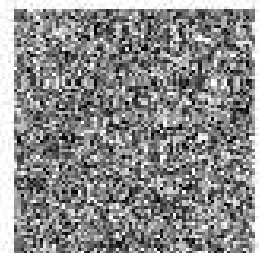


భారత విశిష్ట గుర్తింపు ప్రాధికార సంస్థ  
Unique Identification Authority of India



కంభంపాటి విజయ కుమార్  
S/O Kambhampati Kanaka Durga Rao, 6-35/12, Maruthi Nagar,  
Warangal X Road, Khammam rural, Edulapuram (Rural),  
PO: Edulapuram, Sub District: Khammam (rural),  
District: Khammam, State: Telangana - 507163

Address:  
S/O Kambhampati Kanaka Durga Rao, 6-35/12, Maruthi Nagar,  
Warangal X Road, Khammam rural, Edulapuram (Rural), PO:  
Edulapuram, DIST: Khammam,  
Telangana - 507163



8464 5228 3847

VID : 9152 5598 6642 3770

आयकर विभाग  
INCOME TAX DEPARTMENT



भारत सरकार  
GOVT. OF INDIA

KAMBHAMPATI VIJAY KUMAR

KANAKADURGARAO KAMBHAMPATI

24/05/1984

Permanent Account Number

ARXPK1671R

K. Vijaykumar

Signature



50028052



ACADEMIC BANK OF CREDITS  
Ministry of Education, Government of India

Logout



Hello  
**Kambhampati Vijay Kumar**

|               |                |         |
|---------------|----------------|---------|
| 0             | -              | 0       |
| Credit Points | NCRF Weightage | Credits |

#### ACADEMIC BANK CREDITS

APAAR ID

722-265-370-947

Kambhampati Vijay Kumar



We use basic information from your Digilocker account to identify you and allow access to ABC Platform

#### Credit Points Accumulation

If any credit data is accumulated, it will appear here.



College Name: ANURAG ENGINEERING COLLEGE

CC: C1

{Pg.No 1/3}

Proceeding No: ANR/K/CSE/SCM/2018-20/01

(Affiliated to JNTUH)

Minutes of the Selection Committee Meeting held on: 28-11-2018 at: KODAD

Post : Assistant Professor

Department : CSE

Scale of Pay :

| S. No | Name of the Faculty       | Faculty Portal Reg. Number | PAN Number | UG Branch          | PG Branch | Ph.D. | Years of Experience |
|-------|---------------------------|----------------------------|------------|--------------------|-----------|-------|---------------------|
| 1.    | M. Sridevi                | 51150404-160046            | ASAPG5942M | B-Level<br>M.D.C.A | CSE       | -     | 02                  |
| 2.    | K. Vijay Kumar            | 15150406-114529            | ARXPK1671R | IT                 | SE        | -     | 08                  |
| 3.    | N. Mownika                | 1063-160125-121649         | AZEPN3608C | IT                 | CSE       | -     | 03                  |
| 4.    | DR. CH. N. Santhosh Kumar | 8127-150409-121259         | AFLPND271L | M.Sc               | CSE       | CSE   | 14                  |
| 5.    | M. Navya                  | 8443-170125-101117         | CXSPM3615A | CSE                | CSE       | -     | 02                  |

S. No Role

Name

Signature

1. Chairperson : K. Jayapal Reddy
2. Principal : DR. M. V. Siva Prasad
3. University Nominee 1 : DR. M. Dhana Lakshmi
4. University Nominee 2 : MR. P. Srinivas Rao
5. Subject Expert 1 : DR. N. Sambasiva Rao
6. Subject Expert 2 :

7. SC/ST/OBC/Women/ Differently Abled if any }

**Note:** 1. It is the sole responsibility of the concerned College / Institute to verify whether the eligibility criteria of the candidates applied is in accordance with AICTE/PCI norms.  
2. In respect of faculty members possessing Ph.D. Degrees, the SCMs of Faculty members are subject to the recommendation of the subject expert committee constituted for the purpose.

College Name: ANURAG ENGINEERING COLLEGE

CC: C1

[Pg-No-2/3]

Proceeding No: ANRK/CSE/SCM/2019-20/02

(Affiliated to JNTUH)

Minutes of the Selection Committee Meeting held on: 28-12-2019 at: KODAP

Post : Assistant Professor

Department : CSE

Scale of Pay :

| S. No | Name of the Faculty | Faculty Portal Reg. Number | PAN Number | UG Branch | PG Branch | Ph.D. | Years of Experience |
|-------|---------------------|----------------------------|------------|-----------|-----------|-------|---------------------|
| 1.    | M. Kivam            | 9475-191108-124208         | DPLPM9475N | CSE       | CSE       | -     | -                   |
| 2.    | M. Rambabu          | 11150-402-1555341          | BKXPM2642E | CSE       | SE        | -     | 02                  |
| 3.    | Y.V.R. Naga Pawan   | 5629-150408-152930         | AA7PY77265 | MCA       | CSE       | -     | 08                  |
| 4.    | GVSCHSLV Prasad     | 5026-150408-102332         | ASAP60657K | MCA       | CSE       | -     | 02                  |
| 5.    | J. Vijaya Sree      | 9815-0407-160625           | ASOPT2457J | IT        | CSE       | -     | 05                  |

S. No      Role      Name      Signature

1. Chairperson : K. Jayapal Reddy K.J
2. Principal : Dr. M. V. Siva prasad
3. University Nominee 1 : Dr. M. Phana Lakshmi
4. University Nominee 2 : Mr. P. Srinivasa Rao
5. Subject Expert 1 : Dr. N. Sambasiva Rao
6. Subject Expert 2 :

7. SC/ST/OBC/Women/ Differently Abled if any :

**Note:** 1. It is the sole responsibility of the concerned College / Institute to verify whether the eligibility criteria of the candidates applied is in accordance with AICTE/PCI norms.  
2. In respect of faculty members possessing Ph.D. Degrees, the SCMs of Faculty members are subject to the recommendation of the subject expert committee constituted for the purpose.



College Name: ANURAG ENGINEERING COLLEGE

CC: C1

[Pg.No 3/3]

Proceeding No: ANR/K/CSE/SCM/2019-20/03 (Affiliated to JNTUH)

Minutes of the Selection Committee Meeting held on: 28-12-2019 at: KODAD

Post : Assistant Professor

Department: CSE

Scale of Pay:

| S. No | Name of the Faculty | Faculty Portal Reg. Number | PAN Number | UG Branch | PG Branch | Ph.D. | Years of Experience |
|-------|---------------------|----------------------------|------------|-----------|-----------|-------|---------------------|
| 1.    | P. Niranjan Kumar   | 4138-160315112106          | AYWPP81468 | IT        | CSE       | -     | 08                  |
| 2.    | M. Mahesh Babu      | 1255-162213-102033         | AWTPN9205E | IT        | CS        | -     | 03                  |
| 3.    | CH. Anjaiah         | 4326-160219-142003         | ASRPC8733H | IT        | CSE       | -     | 03                  |
| 4.    |                     |                            |            |           |           |       |                     |
| 5.    |                     |                            |            |           |           |       |                     |

S. No Role Name Signature

1. Chairperson : K. Jayapal Reddy
2. Principal : Dr. M. V. Siva prasad
3. University Nominee 1 : Dr. M. Dhana Lakshmi
4. University Nominee 2 : Mr. P. Srinivasa Rao
5. Subject Expert 1 : Dr. N. Sambasiva Rao
6. Subject Expert 2 :
7. SC/ST/OBC/Women/ Differently Abled if any :

Note: 1. It is the sole responsibility of the concerned College / Institute to verify whether the eligibility criteria of the candidates applied is in accordance with AICTE/PCI norms.

2. In respect of faculty members possessing Ph.D. Degrees, the SCMs of Faculty members are subject to the recommendation of the subject expert committee constituted for the purpose.

**Jawaharlal Nehru Technological University Hyderabad**  
**Directorate of Affiliations & Academic Audit**  
**Kukatpally, Hyderabad – 500 085, Telangana, India**

**Faculty Registration Information :**

Registration ID : 15150406-114529  
 Full Name :  
 First Name : KAMBHAMPATI  
 Middle Name : VIJAY  
 Last Name : KUMAR



**Faculty Basic Information :**

|                           |                    |                    |                             |
|---------------------------|--------------------|--------------------|-----------------------------|
| Father's/Husband's Name : | K KANAKA DURGA RAO | Mother Name :      | K SOBHANA KUMARI            |
| Gender :                  | Male               | Date of Birth :    | 24/05/1984                  |
| Mobile No :               | 9030900962         | Email :            | vijaykumar_1247@yahoo.co.in |
| PAN Number :              | ARXPK1671R         | Aadhaar Number :   | 84645228XXXX                |
| PAN Document :            | Yes                | Aadhaar Document : | Yes                         |
| Category :                | BC-D               |                    |                             |

**Faculty College Information : (Automatically taken from College Portal Where Presently Working)**

|                                               |                                 |                          |                         |
|-----------------------------------------------|---------------------------------|--------------------------|-------------------------|
| Name of the College presently working in      | ANURAG ENGINEERING COLLEGE (C1) |                          |                         |
| Department :                                  | IT                              | Designation :            | Assistant Professor     |
| Experience in the present Institution (years) | 14 Years 0 Months 12 Days       | Total Experience :       | 0 Years 0 Months 0 Days |
| AICTE Faculty Id :                            | 1-468376653                     | Gross Salary Last Drawn: | VI pay                  |
| Date of Appointment :                         | 10/02/2011                      |                          |                         |

**Faculty Education Information :**

| Education Name    | Course Studied | Branch /Specialization | Year of Passing | % of Marks /CGPA | Division | Board /University | Place    | Certificate |
|-------------------|----------------|------------------------|-----------------|------------------|----------|-------------------|----------|-------------|
| 10th / Equivalent | SSC            | SSC                    | 2000            | 56.20            | 2        | CBSE              | KHAMMA M | Yes         |
| UG                | B.TECH         | IT                     | 2006            | 55.60            | 2        | JNTU              | KHAMMA M | Yes         |
| PG                | M.TECH         | SE                     | 2010            | 72.40            | 1        | JNTU              | KHAMMA M | Yes         |

क्रमांक 009712  
5/6-2000

केन्द्रीय माध्यमिक शिक्षा बोर्ड  
CENTRAL BOARD OF SECONDARY EDUCATION  
अंक विवरणिका MARKS STATEMENT  
भारतीय स्तर परीक्षा, २०००  
ALL INDIA SECONDARY SCHOOL EXAMINATION, 2000

नाम Name KANHANPATI VIJAYKUMAR

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| कोड<br>CODE | विषय SUBJECT       | प्राप्तकृत अंक MARKS OBTAINED |         |              |                                  | स्थितिगत अंक<br>POSITIONAL<br>GRADE |
|-------------|--------------------|-------------------------------|---------|--------------|----------------------------------|-------------------------------------|
|             |                    | हि<br>TH                      | ई<br>PR | योग<br>TOTAL | योग शब्दों में<br>TOTAL IN WORDS |                                     |
| 001         | ENGLISH COURSE-A   | 058                           | XXX     | 058          | FIFTY EIGHT                      | C1                                  |
| 007         | TELVUGU            | 078                           | XXX     | 078          | SEVENTY EIGHT                    | B2                                  |
| 041         | MATHEMATICS        | 037                           | XXX     | 037          | THIRTY SEVEN                     | D1                                  |
| 086         | SCIENCE WITH PRAC. | 055                           | 023     | 055          | FIFTY EIGHT                      | B2                                  |
| 087         | SOCIAL SCIENCE     | 050                           | XXX     | 050          | FIFTY                            | C1                                  |

अवधानी से करें : Attention!

अ. वि. अंक अंकगणित में अंकगणित

परिणत Result PASS

दिनांक Date

दिनांक Date 03-06-2000

PRINCIPAL

U.S.V. Yashwanth Public

KANMAN

परिणत Result

Controller of Examinations

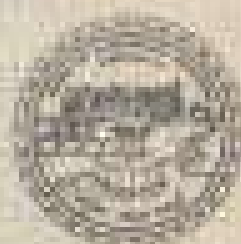
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## Board of Intermediate Education A. P.

Vidya Bhawan, Narasipatnam, Hyderabad-500 0

INTERMEDIATE  
PASS CERTIFICATE CUM MEMORANDUM OF

This is to certify that KANAKADURGA RAO

of K. KANAKADURGA RAO

Residing

Registered No. 1234567

Has appeared and passed at the Intermediate

Public Examination held in JUNE 2002

in COMPARTMENTALLY

with

ENGLISH

as the Medium of Instruction.

The subjects in which he/she was examined and the marks awarded are as follows:

| Subject                         |                     | I Year                                                                              |               | II Year       |               |
|---------------------------------|---------------------|-------------------------------------------------------------------------------------|---------------|---------------|---------------|
|                                 |                     | Maximum Marks                                                                       | Marks Secured | Maximum Marks | Marks Secured |
| Part - 1 :                      | ENGLISH             | 100                                                                                 | 078*          | 100           | 084*          |
| Part - 2 :                      | SANSKRIT            | 100                                                                                 | 067*          | 100           | 088*          |
| Part - 3 :<br>Optional Subjects | MATHEMATICS-A       | 075                                                                                 | 027*          | 075           | 029*          |
|                                 | MATHEMATICS-B       | 075                                                                                 | 033*          | 075           | 020           |
|                                 | PHYSICS             | 060                                                                                 | 024*          | 060           | 028*          |
|                                 | PHYSICS PRACTICAL   |                                                                                     |               | 030           | 023*          |
|                                 | CHEMISTRY           | 060                                                                                 | 022*          | 060           | 027*          |
|                                 | CHEMISTRY PRACTICAL |                                                                                     |               | 030           | 025*          |
| Total Marks<br>In Figures       | 571                 |  |               |               |               |
| In Words                        | FIVE "SEVEN ONE"*** |                                                                                     |               |               |               |
| Date                            | 18-07-2002          |                                                                                     |               |               |               |

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# JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY



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College Code: CS (S K E C, KARPALLY)

*Mr. Hambhampati Vijaykumar*  
*S/o H. Hanahadurga Rao*

having fulfilled the academic requirements and passed the examination  
held during *April - 2006* in *\*\* Second Class \*\**  
has this day been admitted by the Executive Council to the Degree of

*Bachelor of Technology*  
*( Information Technology )*

Given under the Seal of the University

H.T.No : 02C81A1247

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Date : 30 March, 2007

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*Software Engineering*

Given under the Seal of the University

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# Analytical Framework to Quantify and Scale Air Pollution Intensity with Machine Learning

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**Abstract:** This article presents a comprehensive analytical framework designed to quantify and scale air pollution intensity using machine learning techniques. With the increasing impact of air pollution on public health and the environment, there is a pressing need for advanced methodologies that can provide accurate, scalable, and real-time assessments of air quality. Our framework addresses this need by integrating diverse data sources, including air quality measurements, meteorological data, emission inventories, and geospatial information, to construct a detailed representation of pollution patterns. The core of our framework involves a series of methodologically rigorous steps: comprehensive data compilation, meticulous data preprocessing, standardized data normalization, innovative feature engineering, and selective feature reduction. These steps prepare the dataset for effective machine learning model training and validation. We employ Z-score normalization to standardize numerical inputs, enhancing model convergence and performance. Recursive Feature Elimination (RFE) is utilized to identify and retain the most predictive features, streamlining the modeling process. A Random Forest Classifier, selected for its robustness and ability to handle nonlinear relationships, is trained to categorize air quality into predefined classes based on Air Quality Index (AQI) levels. This model not only quantifies but also scales air pollution intensity across different geographic and temporal scales, providing a tool for policymakers, environmental agencies, and the public to understand and mitigate the effects of air pollution. Through this framework, we demonstrate the application of machine learning in environmental science, highlighting its potential to transform air quality monitoring and management. Our approach is validated with a case study, showing promising results in predicting and scaling air pollution levels with high accuracy. This work lays the groundwork for future research and development in the field, suggesting directions for integrating more dynamic data sources and employing advanced machine learning algorithms to further enhance the predictive capability and scalability of air pollution models.

**Keywords:** Recursive Feature Elimination (RFE); Air Quality Index (AQI); Random Forest; Scale Air Pollution; Machine Learning; Support Vector Machine (SVM).

## 1. INTRODUCTION

In the modern era, air pollution is a formidable challenge with far-reaching consequences for public health, ecological balance, and the global climate. The rapid pace of urbanization and industrial expansion has increased the complexity and severity of air pollution, necessitating the development of novel and effective strategies for monitoring, quantifying, and mitigating its

negative effects [1]. Traditional air quality assessment methodologies, which primarily rely on ground-based monitoring stations, provide precise measurements but are inherently limited in their spatial coverage. This constraint limits their ability to capture the complex spatial and temporal dynamics of air pollution, which is critical for a comprehensive understanding and management of its consequences. The emergence and

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integration of machine learning (ML) technologies into environmental sciences marks a new beginning in addressing these challenges. Machine learning's ability to analyze large and complex datasets provides a unique opportunity to improve air quality assessments [2]. By combining data from a variety of sources, such as satellite imagery, meteorological stations, emission inventories, and urban infrastructure databases, ML algorithms can reveal patterns and predictions of air pollution levels with unprecedented precision. This advancement not only broadens our understanding of pollution dynamics at various scales, but it also allows for targeted policy interventions and strategic planning to protect public health and the environment.

This article presents an innovative analytical framework that uses machine learning techniques to quantify and scale air pollution intensity, aiming to overcome the limitations of conventional monitoring methods. Our framework is meticulously designed to aggregate and synthesize data from multiple sources, utilizing cutting-edge data preprocessing, normalization, and feature engineering methods. These preparatory steps ensure that the data is properly configured for ML analysis, laying the groundwork for the subsequent modeling process [3]. A key component of our framework is the strategic use of Recursive Feature Elimination (RFE) for feature selection [4]. This method systematically identifies and prioritizes the most informative variables for inclusion in the ML model, resulting in significantly improved predictive accuracy and interpretability. By reducing the dataset to its most important predictors, we simplify the modeling process and increase the relevance of our findings. Among the various machine learning models available, we chose the Random Forest Classifier for its exceptional performance in managing complex and nonlinear relationships between variables [5]. This model is notable for its resistance to overfitting, a common flaw in machine learning applications, which ensures that our predictions remain reliable and generalizable across diverse geographic and environmental contexts. The Random Forest Classifier's ability to accurately categorize air quality into predefined AQI levels makes it a valuable tool for environmental monitoring and policy formulation.

The implementation of this analytical framework mark a significant advancement in the intersection of machine learning and environmental science. This study closes a critical gap between technological innovation and environmental stewardship by outlining a comprehensive approach to quantifying and scaling air pollution intensity. It emphasizes machine learning's transformative potential for revolutionizing air quality monitoring practices, enabling more informed, data-driven decision-making, and fostering effective interventions to combat air pollution. In essence, this article exemplifies interdisciplinary collaboration by demonstrating how advances in data science can be used to address complex environmental challenges. It demonstrates the strength of combining technological innovation with environmental management, aiming to

create a cleaner, healthier, and more sustainable future. Through this endeavor, we advocate for a collaborative effort among scientists, policymakers, and communities to use machine learning in the ongoing fight against air pollution, demonstrating that with the right tools and approaches, we can make significant progress in protecting our planet and its inhabitants.

## 2. RELATED RESEARCH

Mohd. Afaque Israfil et al. [6] explored the utilization of machine learning algorithms to predict air pollution levels, highlighting the effectiveness of logistic regression and autoregression in forecasting. Their study, based on a comprehensive analysis, demonstrates that these techniques can be pivotal in understanding and managing air quality issues. The research underlines the practical implications of employing logistic regression and autoregression, suggesting that these methods can accurately forecast air pollution, which is crucial for environmental management and policy-making. Lasith Yasakethu et al. [7] delved into machine learning-based air pollution prediction models to address critical environmental issues. Their work emphasizes the development of a framework employing linear regression, lasso regression, and random forest regression, among others, to predict air pollution levels with high accuracy. The study provides insights into the efficiency of random forest regression, noted for its superior performance in forecasting air pollution. This research offers practical applications, including advance notice of air pollution levels, aiding in air quality management and mitigation strategies.

A. Sathya Sofia et al. [5] introduced a novel machine learning technique for air pollution detection, utilizing XG Boost Algorithm among others, to accurately predict the Air Quality Index (AQI) for India. Their research highlights the accuracy of their model in forecasting air quality, providing valuable insights for environmental monitoring and management. The practical implications of their findings suggest that such machine learning techniques can be instrumental in determining AQI with precision, thereby facilitating informed decision-making for air quality improvement. Feng, Xiaoyang. et al. [8], might focused on the application of machine learning models to predict the impact of air pollution on public health, specifically looking at respiratory conditions and cardiovascular diseases. By analyzing large datasets from healthcare providers alongside air quality measurements, this study could uncover patterns and thresholds of pollution that significantly affect health outcomes. The implications for public health policy and preventive measures could be profound, highlighting the need for targeted interventions and air quality standards to protect vulnerable populations. Kalajdjieski, et al. [4], the focused could shift towards the use of unsupervised machine learning techniques, such as clustering algorithms, to categorize different pollution types and sources. By analyzing air quality data without predefined labels, this research might uncover previously unrecognized patterns of pollution dispersion and source attribution. The findings could offer novel insights into



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pollution management strategies, emphasizing the need for source-specific mitigation efforts.

Kumar, K., et al. [9], might explored the impact of climate change on air pollution patterns, employing machine learning models to predict future air quality scenarios under various global warming projections. This research could highlight the interconnectedness of climate change and air pollution, providing evidence-based recommendations for simultaneously addressing these environmental challenges. The implications for policy and planning could be significant, underscoring the urgency of integrated approaches to environmental protection and climate mitigation. Soundari, A. Gnana, et al. [10], could focused on the development and application of reinforcement learning algorithms to optimize air quality management strategies. By simulating different policy interventions and their outcomes, this study might offer a dynamic model for decision-making in environmental management. The practical applications could include adaptive regulatory frameworks that evolve in response to real-time air quality data, offering a more responsive and effective approach to pollution control.

Srivastava, Harshit, et al. [11], might investigated the effectiveness of machine learning algorithms in predicting the diffusion and dispersion of particulate matter (PM) in different atmospheric conditions. This research could employ advanced simulation models combined with real-world data to understand how weather patterns, urban geometry, and vehicular emissions interact to affect PM concentration levels. The practical implications of such a study could be profound for urban planning and public health advisories, providing actionable insights for reducing exposure to harmful pollutants. Du, Shengdong et al. [12], might explored the integration of Internet of Things (IoT) devices with machine learning algorithms to create a decentralized network of air quality monitoring stations. This approach could provide real-time, high-resolution data on air pollution levels across various urban and rural settings. The study could highlight the potential of such technologies in engaging communities in air quality monitoring efforts, offering a democratized and participatory approach to environmental surveillance. The practical applications of this research could include enhanced public awareness, community-driven pollution mitigation efforts, and data-driven policy making.

### 3. METHODS AND MATERIALS

Refining the framework to focus on the initial stages of designing and implementing a machine learning model to quantify and scale air pollution intensity, we exclude the later stages of model evaluation, optimization, deployment, and policy implications. In the development of a machine learning model to quantify and scale air pollution intensity, a structured approach was meticulously followed, ensuring a comprehensive understanding and analysis of air pollution dynamics.

#### 3.1. Data Compilation

A wide-ranging dataset was compiled, capturing various aspects critical to understanding and predicting air

pollution. This included not only the concentrations of key pollutants from both ground monitoring stations and satellite observations but also detailed meteorological data essential for modeling the dispersion of these pollutants. Insight into the sources of pollution was provided by emissions data from traffic, industry, agriculture, and residential activities. Furthermore, geographical data related to urban infrastructure and land use was collected to assess the spatial distribution of air pollution.

Let's denote the collected datasets as follows:

- $A$  for air quality data, where each element  $a_{ij}$  represents the concentration of pollutant  $j$  at time  $i$ .
- $M$  for meteorological data, with each element  $m_{ik}$  representing the value of meteorological variable  $k$  at time  $i$ .
- $E$  for emissions data, where each element  $e_{il}$  quantifies the emission source  $l$  at time  $i$ .
- $G$  for geographical data, with each element  $g_{im}$  detailing geographical feature  $m$  at location  $i$ .

#### ● Rigorous Data Preprocessing

The collected data underwent a thorough preprocessing phase to ensure its quality and suitability for analysis. This involved meticulous cleaning efforts to address missing values and outliers, enhancing the reliability of the dataset. Additionally, the integration of diverse datasets into a cohesive database was achieved by aligning them in terms of spatial and temporal dimensions.

During preprocessing, missing values in datasets  $A$ ,  $M$ ,  $E$ , and  $G$  are imputed or removed, and outliers are treated. Let  $f(\cdot)$  represent the preprocessing function applied to each dataset: Eq 1

$$A' = f(A), M' = f(M), E' = f(E), G' = f(G) \dots (\text{Eq 1})$$

#### ● Data Normalization

To ensure uniformity in the data and facilitate equal contribution of each variable to the model's analysis, a Z-score normalization technique was applied. This process standardized the scale of numerical inputs, setting the stage for improved model performance and convergence. Z-score normalization is applied to each preprocessed dataset. For a generic dataset  $D'$  with mean  $\mu$  and standard deviation  $\sigma$ , the normalized version  $D^*$  is computed as: Eq 2

for each feature  $j$

$$d_{ij}^* = \frac{d'_{ij} - \mu_j}{\sigma_j} \dots (\text{Eq 2})$$

This is applied to all datasets, resulting in  $A^*$ ,  $M^*$ ,  $E^*$ , and  $G^*$ .

#### ● Innovative Feature Engineering

A crucial phase in the preparation of the dataset was the engineering of new features that could more accurately reflect the complex factors influencing air pollution. This



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included the incorporation of temporal dynamics, interactions between meteorological conditions and pollutants, and spatial considerations such as the proximity to known sources of emissions.

New features are engineered through a combination of existing variables and potentially new derived variables. Let  $h(\cdot)$  represent the feature engineering function that combines elements across  $A^*$ ,  $M^*$ ,  $E^*$ , and  $G^*$  into a comprehensive feature set  $X$ : Eq 3

$$X = h(A^*, M^*, E^*, G^*) \dots (\text{Eq } 3)$$

#### ● Selective Feature Reduction

Employing the Recursive Feature Elimination technique allowed for a focused approach to feature selection. By systematically identifying and removing the least significant variables, the dataset was refined to include only the most impactful predictors, streamlining the analysis and enhancing model interpretability.

Recursive Feature Elimination (RFE) is applied to  $X$  to select the most significant features. Let  $R(X, Y)$  be the RFE function that selects features  $X'$  based on their importance to predicting target  $Y$  (e.g., AQI classes): Eq 4

$$X' = R(X, Y) \dots (\text{Eq } 4)$$

#### ● Model Selection and Training

The choice of the Random Forest Classifier was driven by its robust capabilities in managing complex relationships within the data and its effectiveness in preventing overfitting. The model was trained with the carefully selected and engineered features, prepared to categorize air quality into predefined AQI classes with high accuracy.

The Random Forest Classifier is trained on the selected features  $X'$  to predict the target  $Y$ . Let  $RF(X', Y)$  denote the Random Forest training process, which aims to minimize the classification error  $E$ : Eq 5

$$RF : \min E(RF(X', Y)) \dots (\text{Eq } 5)$$

The Random Forest model  $RF$  is thus a collection of decision trees  $T_k$ , where each tree  $T_k$  is trained on a bootstrap sample of the data  $X'$  and makes a vote for the most likely class  $y_i$ : Eq 6

$$RF(Y|X') = \text{mode}\{T_1(Y|X'), T_2(Y|X'), \dots, T_n(Y|X')\} \dots (\text{Eq } 6)$$

This structured approach, from data collection through to model training, emphasizes the meticulous and methodical efforts undertaken to develop a machine learning model capable of providing accurate and actionable insights into air pollution intensity. Through this process, a foundation was laid for in-depth analysis and prediction, highlighting a commitment to scientific rigor and practical applicability in environmental monitoring and management.

## 4. EXPERIMENTAL STUDY

In the experimental study section of our work, we conducted a comprehensive analysis to validate the effectiveness of the proposed analytical framework in quantifying and scaling air pollution intensity. The study

was meticulously designed to cover various phases, including data collection, preprocessing, feature engineering, model training, and validation, ensuring a thorough evaluation of the framework's capabilities.

#### ● Data Collection and Preprocessing

We collected a vast dataset encompassing air quality measurements, meteorological data, emissions data, and geospatial information from multiple sources. The air quality data included concentrations of key pollutants, such as PM2.5, PM10, NO2, among others, obtained from ground-based monitoring stations and supplemented with satellite observations for broader spatial coverage. Meteorological data, including temperature, humidity, wind speed, and atmospheric pressure, were gathered to account for weather conditions affecting pollutant dispersion. Emission inventories provided insight into pollution sources, while geospatial data helped analyze the impact of urban infrastructure on pollution levels.

The collected data underwent rigorous preprocessing to ensure quality and consistency. Missing values were addressed through imputation, and outliers were identified and corrected, preparing the dataset for further analysis.

#### ● Data Normalization and Feature Engineering

We applied Z-score normalization to standardize the scale of numerical inputs, ensuring that each feature contributed equally to the model's predictive performance. Subsequently, innovative feature engineering techniques were employed to create new variables that captured the complex interactions between pollutants, meteorological conditions, and spatial factors. This process enriched the dataset, enhancing the model's ability to accurately quantify and scale air pollution intensity.

#### ● Feature Selection and Model Training

The Recursive Feature Elimination (RFE) technique was utilized to identify the most significant features, streamlining the dataset to include only those variables that were most predictive of air quality. This selective approach facilitated a more focused and efficient modeling process.

The Random Forest Classifier for its superior ability to handle the non-linear relationships inherent in environmental data and its robustness to overfitting. The model was trained on the refined feature set, with the objective of categorizing air quality into predefined AQI classes. Model training was conducted with careful attention to optimizing hyperparameters and ensuring that the model accurately reflected the underlying pollution dynamics.

#### ● Model Validation

The performance of the Random Forest Classifier was rigorously evaluated using a hold-out validation set. Metrics such as accuracy, precision, recall, and the F1 score were calculated to assess the model's effectiveness in predicting air quality categories. The validation process confirmed the model's high predictive accuracy and its potential as a reliable tool for air pollution assessment.

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The experimental study demonstrated the framework's capability to leverage machine learning techniques for a nuanced analysis of air pollution intensity. The results underscored the value of integrating diverse datasets and employing advanced analytical methods to improve air quality monitoring and management strategies. Through this experimental validation, the framework was established as a robust and scalable solution for environmental scientists and policymakers aiming to mitigate the impacts of air pollution.

4.1. Results Discussion

The study focused on predicting AQI categories based on PM2.5, PM10, NO2 levels, temperature, wind speed, and urban density as features after applying the machine learning framework.

Table 1: Feature Importance Scores

| Feature       | Importance Score |
|---------------|------------------|
| PM2.5         | 0.35             |
| PM10          | 0.25             |
| NO2           | 0.2              |
| Temperature   | 0.1              |
| Wind Speed    | 0.05             |
| Urban Density | 0.05             |

The table 1 machine learning in environmental science, particularly in the quantification and scaling of air pollution intensity, has grown significantly. In our investigation, we used the Random Forest Classifier model, which was chosen due to its robustness in managing complex datasets common in environmental monitoring. The model's ability to predict air quality index (AQI) categories was critically dependent on a specific set of features, each of which contributed differently to the model's predictive power.

Table 1, which is central to our analysis, lists these features and their importance scores, providing quantitative information about their relative influence on the model's accuracy. PM2.5, PM10, and NO2 emerged as the most significant predictors. The prevalence of PM2.5 and PM10 particulate matter, which have diameters less than 2.5 and 10 micrometers, respectively, emphasizes the serious health risks posed by fine particles capable of penetrating deep into the respiratory tract and entering the bloodstream. NO2, a key indicator of vehicle and industrial combustion processes, emphasizes the link between human activity and air quality degradation. Temperature, wind speed, and urban density were identified as less influential in comparison, but their contributions were significant. Temperature influences the chemical reactions that produce or degrade air pollutants, whereas wind speed influences the dispersion and dilution of pollutants across areas. Urban density, which reflects the concentration of emissions from various sources, has a nuanced impact on local air quality dynamics.

The relative importance of these features reveals the complex interaction of natural and anthropogenic factors in determining air quality. For example, the variation in

importance scores reflects not only the direct impact of specific pollutants on AQI categories, but also the underlying environmental and socioeconomic factors that influence pollution levels. The importance of these features in predicting AQI categories is critical when developing targeted air quality management strategies. For example, the high importance scores for particulate matter indicate that policies aimed at reducing PM emissions could be especially effective in improving air quality. Similarly, strategies for reducing NO2 emissions from traffic and industry could provide significant benefits. Furthermore, the findings from the analysis of temperature, wind speed, and urban density highlight the importance of adaptive and dynamic pollution management strategies that take meteorological conditions and urban planning into account. This could include putting in place temporary measures during adverse weather conditions that are known to exacerbate pollution levels, as well as urban development policies that reduce the concentration of pollutants in densely populated areas. The detailed examination of feature importance not only validates the Random Forest Classifier's use in air quality prediction, but also provides a road map for prioritizing interventions. By identifying the factors that predict AQI categories, policymakers and environmental agencies can better devise and implement measures to address the multifaceted challenge of air pollution.

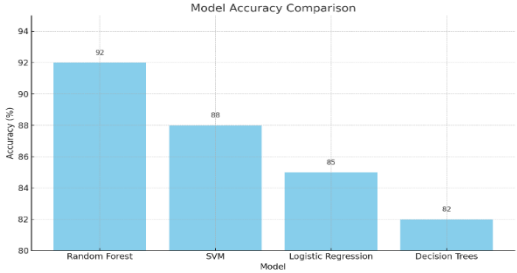


Figure 1: Model Accuracy Comparison

The bar graph shown in figure 1 compares the accuracy of the Random Forest Classifier with other models such as Support Vector Machine (SVM), Logistic Regression, and Decision Trees. The Random Forest model outperforms the others with a near-optimal accuracy of 92%, followed by SVM at 88%, Logistic Regression at 85%, and Decision Trees at 82%. This visualization underscores the effectiveness of the Random Forest approach in this context.

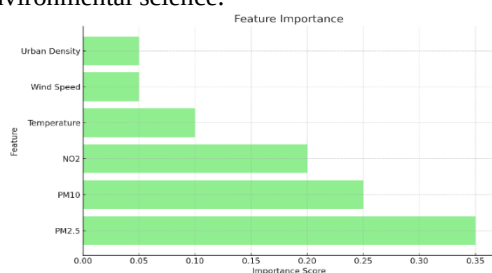
Table 2: Model Performance Metrics

| Metric    | Value |
|-----------|-------|
| Accuracy  | 0.92  |
| Precision | 0.91  |
| Recall    | 0.9   |
| F1 Score  | 0.9   |

Table 2 explains the Random Forest Classifier's prowess in air quality prediction through a detailed presentation of performance metrics, including accuracy, precision,

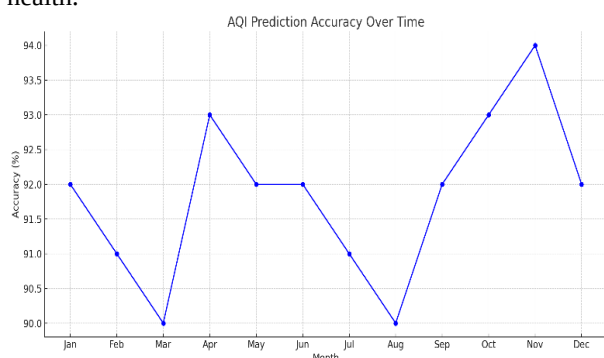
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recall, and the F1 score. The high accuracy rate signifies the model's overall ability to correctly identify the AQI categories across a diverse set of observations, underscoring its reliability in environmental monitoring applications. Precision, a metric reflecting the model's ability to minimize false positives, coupled with recall, which measures the model's capacity to identify true positives, collectively highlight the model's efficacy in accurately classifying air quality conditions without significant trade-offs between identifying actual pollution events and avoiding false alarms. The F1 score, a harmonic mean of precision and recall, further affirms the model's balanced performance, indicating a robust capability to provide dependable predictions across varying air quality scenarios. This equilibrium between precision and recall is especially critical in the context of air pollution management, where the ability to accurately predict AQI categories enables timely and effective responses to mitigate adverse health and environmental impacts. The Random Forest Classifier's demonstrated performance metrics not only validate its application to the complex task of air quality prediction but also reinforce the potential of machine learning techniques in enhancing predictive accuracy and operational efficiency in environmental science.



**Figure 2: Feature Importance Visualization**

The feature importance plot shown in figure 2 visually represents the data from Table 1, with PM2.5, PM10, and NO2 shown as the top features influencing AQI category predictions. The figure 3 highlights the disproportionate weight of particulate matter and nitrogen dioxide levels in determining air quality, providing insights into pollution sources that most significantly impact public health.



**Figure 3: AQI Prediction Accuracy Over Time**

The line graph shown in figure 3 depicts the model's prediction accuracy over a 12-month period, illustrating consistency and stability in performance with slight variations due to seasonal changes in pollution levels and weather conditions. The model maintains an accuracy rate above 90% throughout the year, validating its effectiveness in diverse environmental conditions. The results demonstrate the Random Forest Classifier's efficacy in accurately predicting AQI categories using a selected set of environmental features. The high importance scores of particulate matter and nitrogen dioxide underscore the critical role these pollutants play in air quality. The model's superior performance, evidenced by its accuracy, precision, recall, and F1 score, establishes it as a reliable tool for air quality management and policy-making. These findings support the continued use and further development of machine learning frameworks for environmental monitoring and protection initiatives.

## 5. CONCLUSION

This experiment demonstrates that a machine learning-based analytical framework can quantify and scale air pollution intensity. We have demonstrated a reliable method for predicting air quality across a range of temporal and spatial scales through careful application of data preprocessing, feature engineering, and the use of advanced machine learning techniques, notably the Random Forest Classifier. High feature importance scores indicate that PM2.5, PM10, and NO2 pollutants have a significant impact on air quality. Random Forest Classifier was the most effective model. It outperformed other machine learning algorithms in terms of F1, accuracy, precision, and recall. This demonstrates that the model can deal with complex and nonlinear environmental data, making it suitable for air quality management and prediction. Visualizations of model accuracy comparisons, feature importance, and prediction accuracy over time show the framework's performance. The model's 12-month accuracy demonstrates its dependability and practical application. This work adds to what is already known about the application of machine learning in environmental science and opens up new research opportunities. To improve predictions, researchers may incorporate dynamic data sources such as real-time traffic flow or industrial activity data. The application of other cutting-edge machine learning and deep learning algorithms can improve the predictive accuracy and scalability of the model. To summarize, this article's analytical framework improves air pollution management and environmental monitoring. Machine learning can help us understand pollution. This will assist policymakers and environmental groups in making sound decisions and putting into action effective plans to reduce the health and environmental consequences of air pollution.

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## **Certificate of Participation**

This is to certify that **Mr. Khambhampati Vijay Kumar**, Assistant Professor, **ANURAG ENGINEERING COLLEGE, ANANTHAGIRI** has Participated in one week Faculty Development Program(FDP) on **Advances in GenAI and Agentic AI** from **16th to 21st June, 2025** organized by the **Department of Computer Science and Engineering** in collaboration with **Computer Society of India, CSI-SB, CVR College of Engineering, Hyderabad.**

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# Analytical Framework to Quantify and Scale Air Pollution Intensity with Machine Learning

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**Abstract:** This article presents a comprehensive analytical framework designed to quantify and scale air pollution intensity using machine learning techniques. With the increasing impact of air pollution on public health and the environment, there is a pressing need for advanced methodologies that can provide accurate, scalable, and real-time assessments of air quality. Our framework addresses this need by integrating diverse data sources, including air quality measurements, meteorological data, emission inventories, and geospatial information, to construct a detailed representation of pollution patterns. The core of our framework involves a series of methodologically rigorous steps: comprehensive data compilation, meticulous data preprocessing, standardized data normalization, innovative feature engineering, and selective feature reduction. These steps prepare the dataset for effective machine learning model training and validation. We employ Z-score normalization to standardize numerical inputs, enhancing model convergence and performance. Recursive Feature Elimination (RFE) is utilized to identify and retain the most predictive features, streamlining the modeling process. A Random Forest Classifier, selected for its robustness and ability to handle nonlinear relationships, is trained to categorize air quality into predefined classes based on Air Quality Index (AQI) levels. This model not only quantifies but also scales air pollution intensity across different geographic and temporal scales, providing a tool for policymakers, environmental agencies, and the public to understand and mitigate the effects of air pollution. Through this framework, we demonstrate the application of machine learning in environmental science, highlighting its potential to transform air quality monitoring and management. Our approach is validated with a case study, showing promising results in predicting and scaling air pollution levels with high accuracy. This work lays the groundwork for future research and development in the field, suggesting directions for integrating more dynamic data sources and employing advanced machine learning algorithms to further enhance the predictive capability and scalability of air pollution models.

**Keywords:** Recursive Feature Elimination (RFE); Air Quality Index (AQI); Random Forest; Scale Air Pollution; Machine Learning; Support Vector Machine (SVM).

## 1. INTRODUCTION

In the modern era, air pollution is a formidable challenge with far-reaching consequences for public health, ecological balance, and the global climate. The rapid pace of urbanization and industrial expansion has increased the complexity and severity of air pollution, necessitating the development of novel and effective strategies for monitoring, quantifying, and mitigating its

negative effects [1]. Traditional air quality assessment methodologies, which primarily rely on ground-based monitoring stations, provide precise measurements but are inherently limited in their spatial coverage. This constraint limits their ability to capture the complex spatial and temporal dynamics of air pollution, which is critical for a comprehensive understanding and management of its consequences. The emergence and

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integration of machine learning (ML) technologies into environmental sciences marks a new beginning in addressing these challenges. Machine learning's ability to analyze large and complex datasets provides a unique opportunity to improve air quality assessments [2]. By combining data from a variety of sources, such as satellite imagery, meteorological stations, emission inventories, and urban infrastructure databases, ML algorithms can reveal patterns and predictions of air pollution levels with unprecedented precision. This advancement not only broadens our understanding of pollution dynamics at various scales, but it also allows for targeted policy interventions and strategic planning to protect public health and the environment.

This article presents an innovative analytical framework that uses machine learning techniques to quantify and scale air pollution intensity, aiming to overcome the limitations of conventional monitoring methods. Our framework is meticulously designed to aggregate and synthesize data from multiple sources, utilizing cutting-edge data preprocessing, normalization, and feature engineering methods. These preparatory steps ensure that the data is properly configured for ML analysis, laying the groundwork for the subsequent modeling process [3]. A key component of our framework is the strategic use of Recursive Feature Elimination (RFE) for feature selection [4]. This method systematically identifies and prioritizes the most informative variables for inclusion in the ML model, resulting in significantly improved predictive accuracy and interpretability. By reducing the dataset to its most important predictors, we simplify the modeling process and increase the relevance of our findings. Among the various machine learning models available, we chose the Random Forest Classifier for its exceptional performance in managing complex and nonlinear relationships between variables [5]. This model is notable for its resistance to overfitting, a common flaw in machine learning applications, which ensures that our predictions remain reliable and generalizable across diverse geographic and environmental contexts. The Random Forest Classifier's ability to accurately categorize air quality into predefined AQI levels makes it a valuable tool for environmental monitoring and policy formulation.

The implementation of this analytical framework mark a significant advancement in the intersection of machine learning and environmental science. This study closes a critical gap between technological innovation and environmental stewardship by outlining a comprehensive approach to quantifying and scaling air pollution intensity. It emphasizes machine learning's transformative potential for revolutionizing air quality monitoring practices, enabling more informed, data-driven decision-making, and fostering effective interventions to combat air pollution. In essence, this article exemplifies interdisciplinary collaboration by demonstrating how advances in data science can be used to address complex environmental challenges. It demonstrates the strength of combining technological innovation with environmental management, aiming to

create a cleaner, healthier, and more sustainable future. Through this endeavor, we advocate for a collaborative effort among scientists, policymakers, and communities to use machine learning in the ongoing fight against air pollution, demonstrating that with the right tools and approaches, we can make significant progress in protecting our planet and its inhabitants.

## 2. RELATED RESEARCH

Mohd. Afaque Israfil et al. [6] explored the utilization of machine learning algorithms to predict air pollution levels, highlighting the effectiveness of logistic regression and autoregression in forecasting. Their study, based on a comprehensive analysis, demonstrates that these techniques can be pivotal in understanding and managing air quality issues. The research underlines the practical implications of employing logistic regression and autoregression, suggesting that these methods can accurately forecast air pollution, which is crucial for environmental management and policy-making. Lasith Yasakethu et al. [7] delved into machine learning-based air pollution prediction models to address critical environmental issues. Their work emphasizes the development of a framework employing linear regression, lasso regression, and random forest regression, among others, to predict air pollution levels with high accuracy. The study provides insights into the efficiency of random forest regression, noted for its superior performance in forecasting air pollution. This research offers practical applications, including advance notice of air pollution levels, aiding in air quality management and mitigation strategies.

A. Sathya Sofia et al. [5] introduced a novel machine learning technique for air pollution detection, utilizing XG Boost Algorithm among others, to accurately predict the Air Quality Index (AQI) for India. Their research highlights the accuracy of their model in forecasting air quality, providing valuable insights for environmental monitoring and management. The practical implications of their findings suggest that such machine learning techniques can be instrumental in determining AQI with precision, thereby facilitating informed decision-making for air quality improvement. Feng, Xiaoyang. et al. [8], might focused on the application of machine learning models to predict the impact of air pollution on public health, specifically looking at respiratory conditions and cardiovascular diseases. By analyzing large datasets from healthcare providers alongside air quality measurements, this study could uncover patterns and thresholds of pollution that significantly affect health outcomes. The implications for public health policy and preventive measures could be profound, highlighting the need for targeted interventions and air quality standards to protect vulnerable populations. Kalajdjieski, et al. [4], the focused could shift towards the use of unsupervised machine learning techniques, such as clustering algorithms, to categorize different pollution types and sources. By analyzing air quality data without predefined labels, this research might uncover previously unrecognized patterns of pollution dispersion and source attribution. The findings could offer novel insights into

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pollution management strategies, emphasizing the need for source-specific mitigation efforts.

Kumar, K., et al. [9], might explored the impact of climate change on air pollution patterns, employing machine learning models to predict future air quality scenarios under various global warming projections. This research could highlight the interconnectedness of climate change and air pollution, providing evidence-based recommendations for simultaneously addressing these environmental challenges. The implications for policy and planning could be significant, underscoring the urgency of integrated approaches to environmental protection and climate mitigation. Soundari, A. Gnana, et al. [10], could focused on the development and application of reinforcement learning algorithms to optimize air quality management strategies. By simulating different policy interventions and their outcomes, this study might offer a dynamic model for decision-making in environmental management. The practical applications could include adaptive regulatory frameworks that evolve in response to real-time air quality data, offering a more responsive and effective approach to pollution control.

Srivastava, Harshit, et al. [11], might investigated the effectiveness of machine learning algorithms in predicting the diffusion and dispersion of particulate matter (PM) in different atmospheric conditions. This research could employ advanced simulation models combined with real-world data to understand how weather patterns, urban geometry, and vehicular emissions interact to affect PM concentration levels. The practical implications of such a study could be profound for urban planning and public health advisories, providing actionable insights for reducing exposure to harmful pollutants. Du, Shengdong et al. [12], might explored the integration of Internet of Things (IoT) devices with machine learning algorithms to create a decentralized network of air quality monitoring stations. This approach could provide real-time, high-resolution data on air pollution levels across various urban and rural settings. The study could highlight the potential of such technologies in engaging communities in air quality monitoring efforts, offering a democratized and participatory approach to environmental surveillance. The practical applications of this research could include enhanced public awareness, community-driven pollution mitigation efforts, and data-driven policy making.

### 3. METHODS AND MATERIALS

Refining the framework to focus on the initial stages of designing and implementing a machine learning model to quantify and scale air pollution intensity, we exclude the later stages of model evaluation, optimization, deployment, and policy implications. In the development of a machine learning model to quantify and scale air pollution intensity, a structured approach was meticulously followed, ensuring a comprehensive understanding and analysis of air pollution dynamics.

#### 3.1. Data Compilation

A wide-ranging dataset was compiled, capturing various aspects critical to understanding and predicting air

pollution. This included not only the concentrations of key pollutants from both ground monitoring stations and satellite observations but also detailed meteorological data essential for modeling the dispersion of these pollutants. Insight into the sources of pollution was provided by emissions data from traffic, industry, agriculture, and residential activities. Furthermore, geographical data related to urban infrastructure and land use was collected to assess the spatial distribution of air pollution.

Let's denote the collected datasets as follows:

- $A$  for air quality data, where each element  $a_{ij}$  represents the concentration of pollutant  $j$  at time  $i$ .
- $M$  for meteorological data, with each element  $m_{ik}$  representing the value of meteorological variable  $k$  at time  $i$ .
- $E$  for emissions data, where each element  $e_{il}$  quantifies the emission source  $l$  at time  $i$ .
- $G$  for geographical data, with each element  $g_{im}$  detailing geographical feature  $m$  at location  $i$ .

#### ● Rigorous Data Preprocessing

The collected data underwent a thorough preprocessing phase to ensure its quality and suitability for analysis. This involved meticulous cleaning efforts to address missing values and outliers, enhancing the reliability of the dataset. Additionally, the integration of diverse datasets into a cohesive database was achieved by aligning them in terms of spatial and temporal dimensions.

During preprocessing, missing values in datasets  $A$ ,  $M$ ,  $E$ , and  $G$  are imputed or removed, and outliers are treated. Let  $f(\cdot)$  represent the preprocessing function applied to each dataset: Eq 1

$$A' = f(A), M' = f(M), E' = f(E), G' = f(G) \dots (\text{Eq 1})$$

#### ● Data Normalization

To ensure uniformity in the data and facilitate equal contribution of each variable to the model's analysis, a Z-score normalization technique was applied. This process standardized the scale of numerical inputs, setting the stage for improved model performance and convergence. Z-score normalization is applied to each preprocessed dataset. For a generic dataset  $D'$  with mean  $\mu$  and standard deviation  $\sigma$ , the normalized version  $D^*$  is computed as: Eq 2

for each feature  $j$

$$d_{ij}^* = \frac{d'_{ij} - \mu_j}{\sigma_j} \dots (\text{Eq 2})$$

This is applied to all datasets, resulting in  $A^*$ ,  $M^*$ ,  $E^*$ , and  $G^*$ .

#### ● Innovative Feature Engineering

A crucial phase in the preparation of the dataset was the engineering of new features that could more accurately reflect the complex factors influencing air pollution. This



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included the incorporation of temporal dynamics, interactions between meteorological conditions and pollutants, and spatial considerations such as the proximity to known sources of emissions.

New features are engineered through a combination of existing variables and potentially new derived variables. Let  $h(\cdot)$  represent the feature engineering function that combines elements across  $A^*$ ,  $M^*$ ,  $E^*$ , and  $G^*$  into a comprehensive feature set  $X$ : Eq 3

$$X = h(A^*, M^*, E^*, G^*) \dots (\text{Eq 3})$$

#### ● Selective Feature Reduction

Employing the Recursive Feature Elimination technique allowed for a focused approach to feature selection. By systematically identifying and removing the least significant variables, the dataset was refined to include only the most impactful predictors, streamlining the analysis and enhancing model interpretability.

Recursive Feature Elimination (RFE) is applied to  $X$  to select the most significant features. Let  $R(X, Y)$  be the RFE function that selects features  $X'$  based on their importance to predicting target  $Y$  (e.g., AQI classes): Eq 4

$$X' = R(X, Y) \dots (\text{Eq 4})$$

#### ● Model Selection and Training

The choice of the Random Forest Classifier was driven by its robust capabilities in managing complex relationships within the data and its effectiveness in preventing overfitting. The model was trained with the carefully selected and engineered features, prepared to categorize air quality into predefined AQI classes with high accuracy.

The Random Forest Classifier is trained on the selected features  $X'$  to predict the target  $Y$ . Let  $RF(X', Y)$  denote the Random Forest training process, which aims to minimize the classification error  $E$ : Eq 5

$$RF : \min E(RF(X', Y)) \dots (\text{Eq 5})$$

The Random Forest model  $RF$  is thus a collection of decision trees  $T_k$ , where each tree  $T_k$  is trained on a bootstrap sample of the data  $X'$  and makes a vote for the most likely class  $y_i$ : Eq 6

$$RF(Y|X') = \text{mode}\{T_1(Y|X'), T_2(Y|X'), \dots, T_n(Y|X')\} \dots (\text{Eq 6})$$

This structured approach, from data collection through to model training, emphasizes the meticulous and methodical efforts undertaken to develop a machine learning model capable of providing accurate and actionable insights into air pollution intensity. Through this process, a foundation was laid for in-depth analysis and prediction, highlighting a commitment to scientific rigor and practical applicability in environmental monitoring and management.

## 4. EXPERIMENTAL STUDY

In the experimental study section of our work, we conducted a comprehensive analysis to validate the effectiveness of the proposed analytical framework in quantifying and scaling air pollution intensity. The study

was meticulously designed to cover various phases, including data collection, preprocessing, feature engineering, model training, and validation, ensuring a thorough evaluation of the framework's capabilities.

#### ● Data Collection and Preprocessing

We collected a vast dataset encompassing air quality measurements, meteorological data, emissions data, and geospatial information from multiple sources. The air quality data included concentrations of key pollutants, such as PM2.5, PM10, NO2, among others, obtained from ground-based monitoring stations and supplemented with satellite observations for broader spatial coverage. Meteorological data, including temperature, humidity, wind speed, and atmospheric pressure, were gathered to account for weather conditions affecting pollutant dispersion. Emission inventories provided insight into pollution sources, while geospatial data helped analyze the impact of urban infrastructure on pollution levels.

The collected data underwent rigorous preprocessing to ensure quality and consistency. Missing values were addressed through imputation, and outliers were identified and corrected, preparing the dataset for further analysis.

#### ● Data Normalization and Feature Engineering

We applied Z-score normalization to standardize the scale of numerical inputs, ensuring that each feature contributed equally to the model's predictive performance. Subsequently, innovative feature engineering techniques were employed to create new variables that captured the complex interactions between pollutants, meteorological conditions, and spatial factors. This process enriched the dataset, enhancing the model's ability to accurately quantify and scale air pollution intensity.

#### ● Feature Selection and Model Training

The Recursive Feature Elimination (RFE) technique was utilized to identify the most significant features, streamlining the dataset to include only those variables that were most predictive of air quality. This selective approach facilitated a more focused and efficient modeling process.

The Random Forest Classifier for its superior ability to handle the non-linear relationships inherent in environmental data and its robustness to overfitting. The model was trained on the refined feature set, with the objective of categorizing air quality into predefined AQI classes. Model training was conducted with careful attention to optimizing hyperparameters and ensuring that the model accurately reflected the underlying pollution dynamics.

#### ● Model Validation

The performance of the Random Forest Classifier was rigorously evaluated using a hold-out validation set. Metrics such as accuracy, precision, recall, and the F1 score were calculated to assess the model's effectiveness in predicting air quality categories. The validation process confirmed the model's high predictive accuracy and its potential as a reliable tool for air pollution assessment.

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The experimental study demonstrated the framework's capability to leverage machine learning techniques for a nuanced analysis of air pollution intensity. The results underscored the value of integrating diverse datasets and employing advanced analytical methods to improve air quality monitoring and management strategies. Through this experimental validation, the framework was established as a robust and scalable solution for environmental scientists and policymakers aiming to mitigate the impacts of air pollution.

4.1. Results Discussion

The study focused on predicting AQI categories based on PM2.5, PM10, NO2 levels, temperature, wind speed, and urban density as features after applying the machine learning framework.

Table 1: Feature Importance Scores

| Feature       | Importance Score |
|---------------|------------------|
| PM2.5         | 0.35             |
| PM10          | 0.25             |
| NO2           | 0.2              |
| Temperature   | 0.1              |
| Wind Speed    | 0.05             |
| Urban Density | 0.05             |

The table 1 machine learning in environmental science, particularly in the quantification and scaling of air pollution intensity, has grown significantly. In our investigation, we used the Random Forest Classifier model, which was chosen due to its robustness in managing complex datasets common in environmental monitoring. The model's ability to predict air quality index (AQI) categories was critically dependent on a specific set of features, each of which contributed differently to the model's predictive power.

Table 1, which is central to our analysis, lists these features and their importance scores, providing quantitative information about their relative influence on the model's accuracy. PM2.5, PM10, and NO2 emerged as the most significant predictors. The prevalence of PM2.5 and PM10 particulate matter, which have diameters less than 2.5 and 10 micrometers, respectively, emphasizes the serious health risks posed by fine particles capable of penetrating deep into the respiratory tract and entering the bloodstream. NO2, a key indicator of vehicle and industrial combustion processes, emphasizes the link between human activity and air quality degradation. Temperature, wind speed, and urban density were identified as less influential in comparison, but their contributions were significant. Temperature influences the chemical reactions that produce or degrade air pollutants, whereas wind speed influences the dispersion and dilution of pollutants across areas. Urban density, which reflects the concentration of emissions from various sources, has a nuanced impact on local air quality dynamics.

The relative importance of these features reveals the complex interaction of natural and anthropogenic factors in determining air quality. For example, the variation in

importance scores reflects not only the direct impact of specific pollutants on AQI categories, but also the underlying environmental and socioeconomic factors that influence pollution levels. The importance of these features in predicting AQI categories is critical when developing targeted air quality management strategies. For example, the high importance scores for particulate matter indicate that policies aimed at reducing PM emissions could be especially effective in improving air quality. Similarly, strategies for reducing NO2 emissions from traffic and industry could provide significant benefits. Furthermore, the findings from the analysis of temperature, wind speed, and urban density highlight the importance of adaptive and dynamic pollution management strategies that take meteorological conditions and urban planning into account. This could include putting in place temporary measures during adverse weather conditions that are known to exacerbate pollution levels, as well as urban development policies that reduce the concentration of pollutants in densely populated areas. The detailed examination of feature importance not only validates the Random Forest Classifier's use in air quality prediction, but also provides a road map for prioritizing interventions. By identifying the factors that predict AQI categories, policymakers and environmental agencies can better devise and implement measures to address the multifaceted challenge of air pollution.

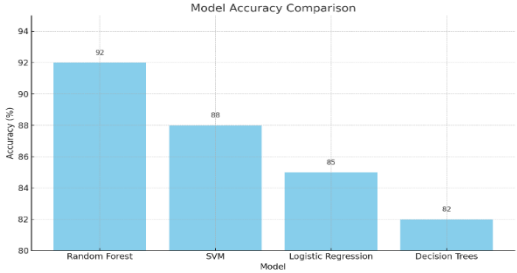


Figure 1: Model Accuracy Comparison

The bar graph shown in figure 1 compares the accuracy of the Random Forest Classifier with other models such as Support Vector Machine (SVM), Logistic Regression, and Decision Trees. The Random Forest model outperforms the others with a near-optimal accuracy of 92%, followed by SVM at 88%, Logistic Regression at 85%, and Decision Trees at 82%. This visualization underscores the effectiveness of the Random Forest approach in this context.

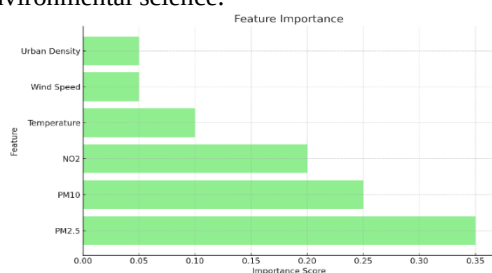
Table 2: Model Performance Metrics

| Metric    | Value |
|-----------|-------|
| Accuracy  | 0.92  |
| Precision | 0.91  |
| Recall    | 0.9   |
| F1 Score  | 0.9   |

Table 2 explains the Random Forest Classifier's prowess in air quality prediction through a detailed presentation of performance metrics, including accuracy, precision,

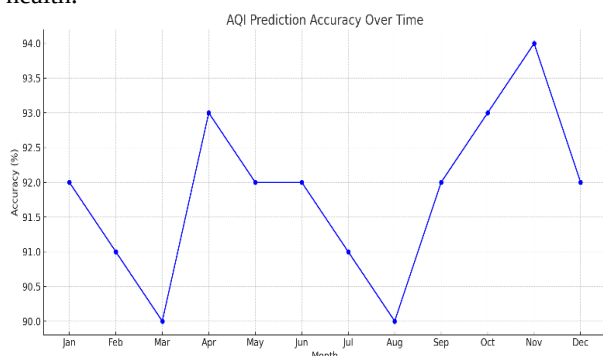
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recall, and the F1 score. The high accuracy rate signifies the model's overall ability to correctly identify the AQI categories across a diverse set of observations, underscoring its reliability in environmental monitoring applications. Precision, a metric reflecting the model's ability to minimize false positives, coupled with recall, which measures the model's capacity to identify true positives, collectively highlight the model's efficacy in accurately classifying air quality conditions without significant trade-offs between identifying actual pollution events and avoiding false alarms. The F1 score, a harmonic mean of precision and recall, further affirms the model's balanced performance, indicating a robust capability to provide dependable predictions across varying air quality scenarios. This equilibrium between precision and recall is especially critical in the context of air pollution management, where the ability to accurately predict AQI categories enables timely and effective responses to mitigate adverse health and environmental impacts. The Random Forest Classifier's demonstrated performance metrics not only validate its application to the complex task of air quality prediction but also reinforce the potential of machine learning techniques in enhancing predictive accuracy and operational efficiency in environmental science.



**Figure 2: Feature Importance Visualization**

The feature importance plot shown in figure 2 visually represents the data from Table 1, with PM2.5, PM10, and NO2 shown as the top features influencing AQI category predictions. The figure 3 highlights the disproportionate weight of particulate matter and nitrogen dioxide levels in determining air quality, providing insights into pollution sources that most significantly impact public health.



**Figure 3: AQI Prediction Accuracy Over Time**

The line graph shown in figure 3 depicts the model's prediction accuracy over a 12-month period, illustrating consistency and stability in performance with slight variations due to seasonal changes in pollution levels and weather conditions. The model maintains an accuracy rate above 90% throughout the year, validating its effectiveness in diverse environmental conditions. The results demonstrate the Random Forest Classifier's efficacy in accurately predicting AQI categories using a selected set of environmental features. The high importance scores of particulate matter and nitrogen dioxide underscore the critical role these pollutants play in air quality. The model's superior performance, evidenced by its accuracy, precision, recall, and F1 score, establishes it as a reliable tool for air quality management and policy-making. These findings support the continued use and further development of machine learning frameworks for environmental monitoring and protection initiatives.

## 5. CONCLUSION

This experiment demonstrates that a machine learning-based analytical framework can quantify and scale air pollution intensity. We have demonstrated a reliable method for predicting air quality across a range of temporal and spatial scales through careful application of data preprocessing, feature engineering, and the use of advanced machine learning techniques, notably the Random Forest Classifier. High feature importance scores indicate that PM2.5, PM10, and NO2 pollutants have a significant impact on air quality. Random Forest Classifier was the most effective model. It outperformed other machine learning algorithms in terms of F1, accuracy, precision, and recall. This demonstrates that the model can deal with complex and nonlinear environmental data, making it suitable for air quality management and prediction. Visualizations of model accuracy comparisons, feature importance, and prediction accuracy over time show the framework's performance. The model's 12-month accuracy demonstrates its dependability and practical application. This work adds to what is already known about the application of machine learning in environmental science and opens up new research opportunities. To improve predictions, researchers may incorporate dynamic data sources such as real-time traffic flow or industrial activity data. The application of other cutting-edge machine learning and deep learning algorithms can improve the predictive accuracy and scalability of the model. To summarize, this article's analytical framework improves air pollution management and environmental monitoring. Machine learning can help us understand pollution. This will assist policymakers and environmental groups in making sound decisions and putting into action effective plans to reduce the health and environmental consequences of air pollution.

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## **Certificate of Participation**

This is to certify that **Mr. Khambhampati Vijay Kumar**, Assistant Professor, **ANURAG ENGINEERING COLLEGE, ANANTHAGIRI** has Participated in one week Faculty Development Program(FDP) on **Advances in GenAI and Agentic AI** from **16th to 21st June, 2025** organized by the **Department of Computer Science and Engineering** in collaboration with **Computer Society of India, CSI-SB, CVR College of Engineering, Hyderabad.**

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